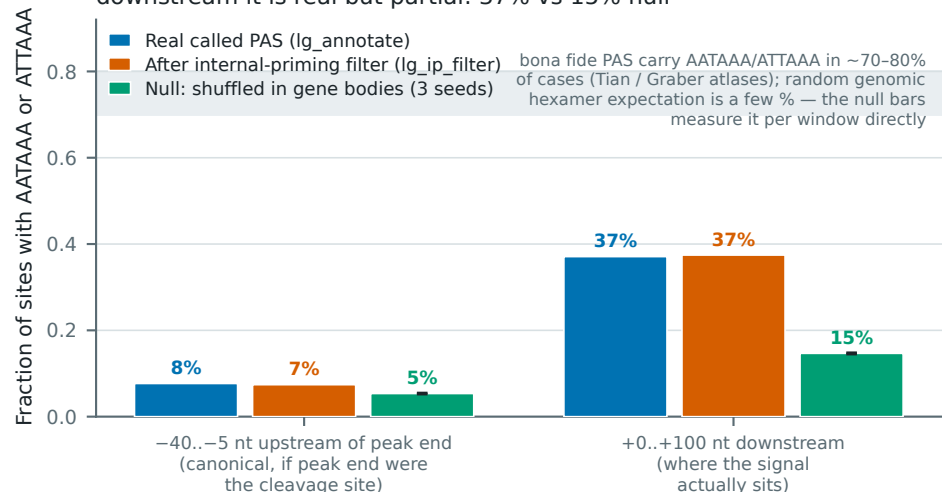
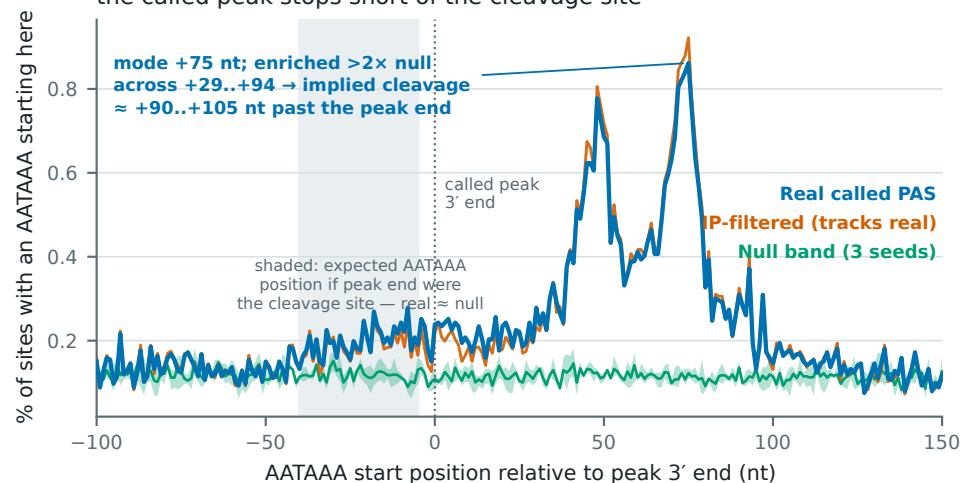


Sequence, with zero atlas input: called peaks flank genuine poly(A) sites, but their 3' ends stop ~90-105 nt short of the cleavage site

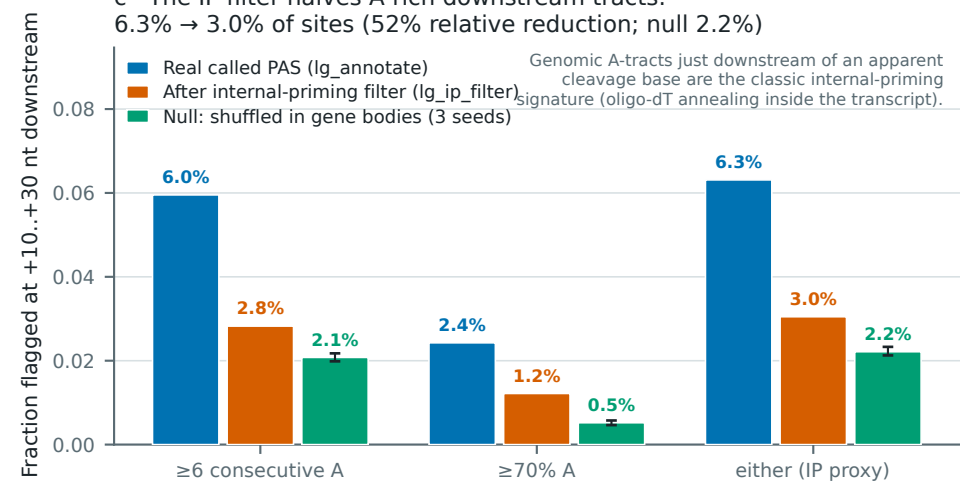
a Upstream of the peak end the signal is near-null (8% vs 5%); downstream it is real but partial: 37% vs 15% null



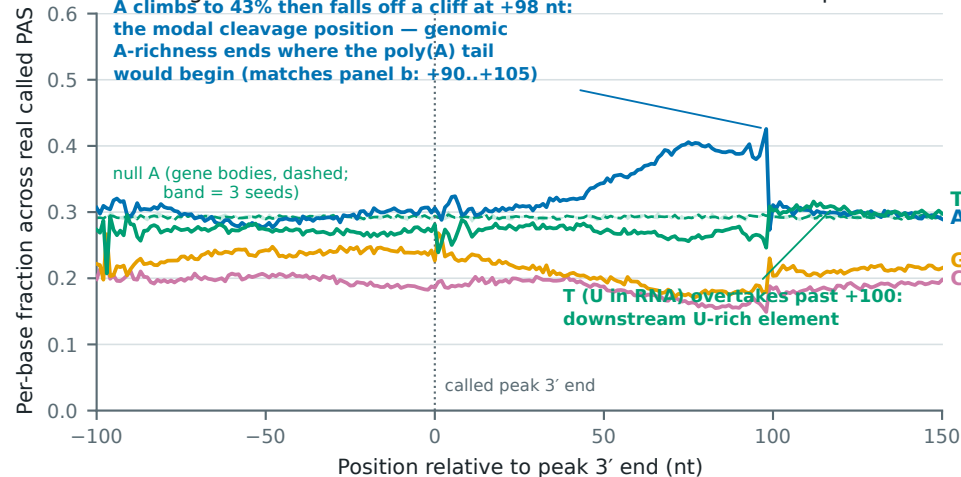
b The AATAAA hump sits downstream of the peak end (mode +75 nt) — the called peak stops short of the cleavage site



c The IP filter halves A-rich downstream tracts: 6.3% → 3.0% of sites (52% relative reduction; null 2.2%)



d Base composition shows the whole PAS architecture — A-rich signal then U-rich element — shifted downstream of the peak end



Anchor r=0 is each peak's 3'-most base per strand (pasbed.bed carries no finer cleavage estimate); windows are strand-aware genomic sequence, bedtools getfasta -s on symmetric ±160 nt windows (GRCh38 primary assembly, Ensembl.99 naming). Sets: real = lg_annotate (n=22,629; 4 scaffold + 0 chrom-edge sites dropped), IP-filtered = lg_ip_filter (n=20,605), null = real intervals shuffled within merged gene bodies on the same chromosome (bedtools shuffle -chrom -noOverlapping -incl, 3 seeds, same n and widths), all collapsed the same way. Hexamers must sit fully inside the stated window; with all 12 canonical variants the fractions are 20% / 19% (real/null, upstream window) and 66% / 44% (+0..+100; single-hexamer noise is high in a 101-nt window). The two windows differ in width, so their null expectations differ — compare each bar with its own null. Implied cleavage offset assumes the canonical AATAAA-to-cleavage spacing of 15-30 nt; the displacement is consistent with 10x R2 coverage ending before the poly(A) junction. In the originally prescribed profile range (-60..+10) the real maximum (-8 nt, 0.28%) barely clears the null band (0.17%); the downstream hump is bimodal (sub-peaks ~+48 and ~+75 nt), suggesting at least two offset populations. Even at +0..+100 the AATAAA/ATATAA rate (37%) stays below the 70-80% of curated PAS — offset variability and residual false calls are not separable here. PolyASite is not used anywhere in this figure. All numbers: motif_validation.tsv.